



Airplane Scatter Plugin

Quick Reference · FM-DX-Webserver · Author: Highpoint

v2.1 · April 2026

✈️ WHAT IS AIRPLANE SCATTER?

An aircraft at **8 000–12 000 m cruise altitude** reflects FM signals between a distant transmitter and your receiver — even when both are beyond each other's radio horizon.

The plugin **predicts these events in real time**, up to 3 min ahead, and displays them on an interactive map with a live countdown.

Best path lengths: **400–1 500 km** · Typical event duration: **15–60 s**

⚙️ HOW IT WORKS

- TX database served from **tx_search.js RAM** (no redundant download) **2.1**
- Fetches live ADS-B positions every **15 seconds**
adsb.one → adsb.lol → adsb.fi (auto-failover)
Large radius split into hexagonal tiles; 500 ms delay between tiles
- Evaluates scatter geometry for every **aircraft × TX pair**
- Applies radio-horizon, ellipse & elevation-angle gates
- Ranks results by **composite score 0–100 %**
- Shows top candidates with **ETA countdown** on map
- All external API calls routed via **local Node proxy** **NEW**

📊 SCATTER SCORE (0–100 %)

Factor	Weight
Cross-track distance ($\sigma = 25$ km)	30 pts
Reflection angle (optimum 40° , $\sigma = 25^\circ$)	25 pts
Aircraft altitude (1 500–12 000 m)	15 pts
Fuselage alignment ($\perp$ bisector)	10 pts
TX ERP (logarithmic)	10 pts
FM frequency (87.5 vs 108 MHz)	5 pts

× Size multiplier: A5 heavy × **1.30** · A4 large × 1.15 · A3 × 1.00 · A2 × 0.85 · A1 × 0.70

Red ≥ 80 % Excellent **Orange** ≥ 60 % High
Yellow ≥ 40 % Medium **Green** < 40 % Low

🕒 ETA COUNTDOWN

Candidates appear **3 min before** and remain **1 min after** the crossing point.

–2:45 Approaching (blue)

NOW ✓ At crossing ±5 s (green)

+0:38 Past crossing (red)

Dead reckoning fills the gap between 15-second ADS-B updates — icon & ETA refresh *every second* smoothly.

Score forecast evaluates **11 time steps** (0 – 300 s ahead); best score is used.

📡 DATA SOURCES

- **TX DB:** tx_search.js RAM **2.1**
freq, ERP, city, country
- **ADS-B:** adsb.one / .lol / .fi
- **Elevation:** opentopodata.org
SRTM 90 m, batch, cached
- **Audio streams:** api.fmlist.org
NEW
- **Flags:** flagcdn.com (ITU→ISO)

📏 GEOMETRY GATES

- TX–RX ≥ **400 km**
- Elevation angle to AC < **9°** (TX & RX)
- Both ends inside aircraft's **radio horizon**
- $d_{tx} + d_{rx} \leq 1.02\text{--}1.08 \times d_{direct}$
- Along-track $\leq d_{txrx} \times 1.1$

🗺️ MAP PANEL

- ● Receiver QTH dot
- ● ● TX dots (sized by ERP)
- ✈️ Aircraft icon (rotated, colour = score)
- - - - Scatter path TX ↔ AC ↔ RX
- Left: candidate list sorted by |ETA|

- **Rotator/GPS:** FM-DX-Webserver WebSocket

- **Proxy:** local

`/api/airplanescatter/proxy`

NEW

- Aircraft > 50 kts & > 1 000 ft

- Bottom: terrain profile (interactive) 2.1
- ► Audio stream Play/Stop per freq NEW
- Status bar: AC count · candidates · DB size

NEW NEW & CHANGED IN V2.1

- **TX DB from RAM** 2.1 — TX data now taken directly from `tx_search.js` in-process RAM; the separate download from maps.fmdx.org is no longer needed
- **Sweet Spot label fix** 2.1 — label shifted +14 px downward so it appears below the midpoint of the green band instead of overlapping its upper edge
- **ERP migration bug resolved** 2.1 — sporadic incorrect ERP values after database migration corrected
- **Elevation profile hover tooltip** 2.1 — new `initProfileCanvasHover()`; mouse-over the terrain canvas shows:
 - Distance from RX / TX at cursor
 - Terrain height at cursor (interpolated)
 - Elevation angle from RX to terrain ●
 - Elevation angle from TX to terrain ●
 - Per-aircraft el. angles from RX & TX (if AC within ±30 km)

v2.0 features below ↓

- **Local Node Proxy** (`airplanescatter_server.js`): replaces public CORS proxy — domain whitelist, caller guard, response size caps
- **Audio Streaming:** Play/Stop live web radio for any TX station directly in the browser (highest bitrate auto-selected)
- **Aircraft Category Filter:** hide light planes & helicopters (A3+ Large & above)
- **Frequency Blacklist / Whitelist:** load static `blacklist.txt` / `whitelist.txt` from plugin folder
- **ADS-B tile fetching:** large radius split into overlapping hexagonal tiles; 500 ms delay between tiles prevents rate-limiting
- **Min score default** lowered: **75 %** → **70 %**
- **GPS & PST Rotator** added as recommended companion plugins

💡 TIPS FOR BEST RESULTS

- **Hover tooltip:** move the mouse over the elevation profile to read exact terrain height, elevation angles from both ends, and per-aircraft angles — helps verify whether the plane is actually inside the sweet spot 2.1
- **Audio Streaming:** start the web radio ~30 s before crossing — streams are typically 15–45 s delayed; compare the burst on air with the stream to positively ID the station NEW
- Use **AI Denoiser** for weak scatter audio github.com/Highpoint2000/AI-Denoise
- Use **RDS AI Decoder** for fragmented RDS bursts github.com/Highpoint2000/RDS-AI-Decoder
- Heavy jets (A380, B747) = **+30 % score bonus** — prioritise!
- Check the green **sweet spot** band in the elevation profile — if the aircraft dot is not inside it, angles are suboptimal
- Low-power hardware: reduce Fetch Radius to 500 km, raise Min ERP to 100 kW
- Click ↻ **Reload** to force a fresh TX database download

⚙️ KEY USER SETTINGS (VIA ⚙️ BUTTON IN PANEL HEADER)

Setting	Default	What it does
Min TX–RX Distance	400 km	Excludes nearby TX. Scatter needs both ends below radio horizon.

Min TX ERP	100 kW	Filters weak stations. Lower = more candidates, more CPU.
Lead / Trail Time	3:00 / 1:00	Window around crossing during which candidate is displayed.
Fetch Radius	750 km	Radius for TX database & ADS-B aircraft request.
Min Scatter Score	70 %	Minimum peak score required to create a candidate entry.
Aircraft Type Filter NEW	All aircraft	Filter out small aircraft / rotorcraft by ADS-B category (e.g. A3+ Large & above).
Frequency Filter List NEW	None	Load a blacklist.txt or whitelist.txt from the plugin folder to suppress or exclusively show specific frequencies.
RX Antenna Height (AGL)	10 m	Your antenna height above ground — affects horizon & profile.
Unit System	Metric	Switch between km/m and miles/ft/knots.

🔧 RECOMMENDED COMPANION PLUGINS

GPS Plugin — provides live moving QTH coordinates via `/data_plugins` WebSocket. Essential for mobile DXing.
github.com/Highpoint2000/GPS

PST Rotator Plugin — enables Rotor Sync Lock & click-to-turn azimuth. Antenna follows scatter candidates automatically.
github.com/Highpoint2000/PSTRotator

AI Denoiser — neural-network (DTLN/RNNoise) real-time audio cleanup for buried scatter signals.
github.com/Highpoint2000/AI-Denoise

RDS AI Decoder — reconstructs PS station name from just a few corrupted RDS bursts using statistical voting & fmdx.org DB.
github.com/Highpoint2000/RDS-AI-Decoder